

CO1-AT1-CODING CHALLENGE

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SLOT: B

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COURSE CODE: CSA0911

QUESTION-6:

CODE:

```
public abstract class Shape {  
    abstract double area();  
  
    public static void main(String[] args) {  
        Shape[] s = {  
            new Circle(5),  
            new Rectangle(4, 6),  
            new Triangle(3, 8)  
        };  
  
        for (Shape x : s) {  
            System.out.println("Area = " + x.area());  
        }  
    }  
}  
  
class Circle extends Shape {  
    double r;  
  
    Circle(double r) {  
        this.r = r;  
    }  
  
    @Override  
    double area() {
```

```
        return 3.14 * r * r;
    }
}

class Rectangle extends Shape {
    double l, b;

    Rectangle(double l, double b) {
        this.l = l;
        this.b = b;
    }

    @Override
    double area() {
        return l * b;
    }
}

class Triangle extends Shape {
    double b, h;

    Triangle(double b, double h) {
        this.b = b;
        this.h = h;
    }

    @Override
    double area() {
        return 0.5 * b * h;
    }
}
```

OUTPUT:

Circle.java

Share

Run

```
1 public abstract class Shape {
2
3     abstract double area();
4
5     public static void main(String[] args) {
6
7         Shape[] s = {
8             new Circle(5),
9             new Rectangle(4, 6),
10            new Triangle(3, 8)
11        };
12
13        for (Shape x : s) {
14            System.out.println("Area = " + x.area());
15        }
16    }
17 }
```

Output

Clear

Area = 78.5
Area = 24.0
Area = 12.0

=== Code Execution Successful ===